

SeedHammer II — the Wallet Policy journey

Engraving a wallet *someone else* built, and proving which one it is.

Three of this repo's journeys follow an operator who owns the seed. This one does not. The wallet arrives as a stack of cards; the operator's whole task is to satisfy themselves that the cards are *the right wallet* before committing them to steel — and the device has to make that possible without ever seeing a secret.

The policy is a depth-2 taproot script tree:

```
tr(@0/<0;1>/*,{pk(@1/<0;1>/*),pk(@2/<0;1>/*)},pk(@3/<0;1>/*))
```

Three properties of it are load-bearing. A taproot Merkle root depends on the tree's **shape**, so no flat descriptor can express this and the address can only be reached through the tree; the four cosigner keys are what push the card set to **8 md1 chunks**, making the gather a real multi-card gather rather than one tap; and four *distinct* keys mean a wrong key-to-slot mapping cannot accidentally produce the right address.

The keys are BIP-39's published test mnemonic ("abandon ... about") at 48' / 0' / N' / 2' . Never put funds behind them.

1 · The host builds the card set

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md encode tr(@0/<0;1>/*,{pk(@1/<0;1>/*),pk(@2/<0;1>/*)},pk(@3/<0;1>/*)}) --key @0=xpub6DkFAXWQ2dHxq2vatrt9qyA3bXYU4ToWQwChbf5XB2mSTexCHZCeKS1VZYcPoBd5X8yVcbXFHJR9R8UCVpt82VX1VhR28mCyxUFL4r6KFrF --key @1=xpub6DzhYrFFYQ1HiMDiM388xHnDiRPNdZJFBmmxge3Y1WwChLTmJLfRuhRHqnQCPbTj3fGKTuKFLHzzwpJkp5Dtc3UtlKZKaVZelyqMBXdx6V --key @2=xpub6EGx8sPr9FxpPE1rbZazhQWpMXA3Hf5DYKtZbL7c4BSddzmQktP96UaTvecEkoCZysuaj79GMCFZYT1KKk7Ph2M3Kf5g8B82KZ8T29SKQR --key @3=xpub6E6Z3S5TXJYNJp4U1q3NZ3pCn82i7KXQAKUtNnzLJ3cCdchQeSdFvXemizaHUF7wNwRQAB8mPdoZhGHLiv49cWPtCnoJY3Az3E8JKxH9Mq --fingerprint @0=73c5da0a --fingerprint @1=73c5da0a --fingerprint @2=73c5da0a --fingerprint @3=73c5da0a --path 48'/0'/0'/2' --force-chunked
chunk-set-id: 0x66289
md1fv c5frs pq6tv yyy4q qxq3z 3ff9g 4sc3q w0za5 zju79 mg9fe utks2 mnchd q59jp g9m5q n4qcn nhfk9 w2m
md1fv c5frs w8egt q4pcw l6u5p 2us6r 6zsnl 2rd0q 6gghv algy0 7r0ck 4wccz rgalt 7lhxu wg56z xata4 9zn
md1fv c5frs ks7d5 e8l2x m3y3x zfmad fj6e2 0ur0a nz7jw kzae8 efvev rz28l lsnr4 qya3s 9ww3k 2y9vw nrm
md1fv c5frs uv7l6 8w6hz ragnj 3g5qr l85ze ape8w q0vdc zfyys 5tqsd 5576t rsa3p 40nfq ms7yz kg0zk 0rc
md1fv c5fr3 zma0q fygnu e7d40 v4943 v82uj lhr4a m7jn7 q2pvh ex36q vgp5c smqq2 9aj7y ay0mm lu38h 9z6
md1fv c5fr3 0eapl franq nlgrl y8y3f gzgre ezipv 7f2n6 9wex2 r0yiw rn79g an0x8 xdw3y smyq0 53dml n56
md1fv c5fr3 k8d59 k03eu 6d7x7 nza0y nr9p5 d3exa n7jvu p3kx8 742q6 d662q zwx3x qg6rc at2hq j4a5w 88g
md1fv c5fr3 7ug5x lcrjj mrdtr jymyw 39w3d 0we0n mjzza 0fvlg tggj3 rhm0l qwpdw nwxqs y7gzl yl246 u0c
note: stdout is watch-only – public keys only, cannot spend
[exit 0]
```

The operator carries 8 md1 chunks.

2 · ...and, separately, what the device must prove to

These come from the *cards*, not from the arguments that produced them — `md inspect` reads the identity the engraved strings will actually carry.

```
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md inspect md1fv c5frs pq6tv yyy4q qxq3z 3ff9g 4sc3q w0za5 zju79 mg9fe utks2 mnchd q59jp g9m5q n4qcn nhfk9 w2m md1fv c5frs w8egt q4pcw l6u5p 2us6r 6zsnl 2rd0q 6gghv algy0 7r0ck 4wccz rga lt 7lhxu wg56z xata4 9zn md1fv c5frs ks7d5 e8l2x m3y3x zfmad fj6e2 0ur0a nz7jw kzae8 efvev rz28l lsnr4 qya3s 9ww3k 2y9vw nrm md1fv c5frs uv7l6 8w6hz ragnj 3g5qr l85ze ape8w q0vdc zfyys 5tqsd 5576t rsa3p 40nfq ms7yz kg0zk 0rc md1fv c5fr3 zma0q fygnu e7d40 v4943 v82uj lhr4a m7jn7 q2pvh ex36q vgp5c smqq2 9aj7y ay0mm lu38h 9z6 md1fv c5fr3 0eapl franq nlgrl y8y3f gzgre ezipv 7f2n6 9wex2 r0yiw rn79g an0x8 xdw3y smyq0 53dml n56 md1fv c5fr3 k8d59 k03eu 6d7x7 nza0y nr9p5 d3exa n7jvu p3kx8 742q6 d662q zwx3x qg6rc at2hq j4a5w 88g md1fv c5fr3 7ug5x lcrjj mrdtr jymyw 39w3d 0we0n mjzza 0fvlg tggj3 rhm0l qwpdw nwxqs y7gzl yl246 u0c
template: tr(@0/<0;1>/*,{pk(@1/<0;1>/*),pk(@2/<0;1>/*)},pk(@3/<0;1>/*)})
n: 4
wallet-policy-mode: true
md1-encoding-id: 6628958a94056d42e86c415960a51acc
wallet-descriptor-template-id: 6f58bdf70d6f7d2fa82e8604bdb89317
wallet-policy-id: ade5967c4b3629f43e714498f426582e
wallet-policy-id-fingerprint: 0xade5967c
note: stdout is a keyless descriptor template (no keys)
[exit 0]

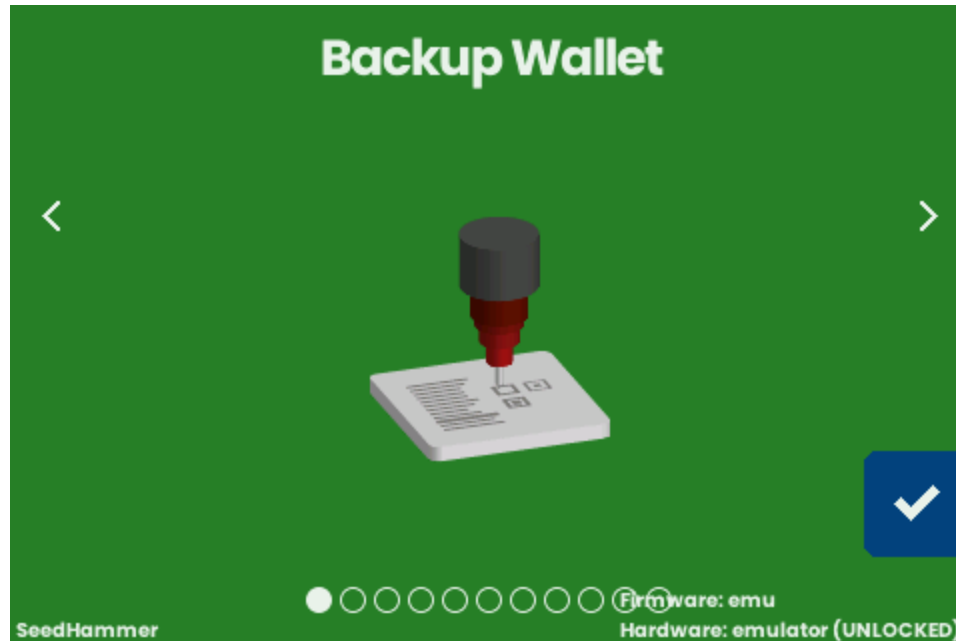
$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md address --template tr(@0/<0;1>/*,{pk(@1/<0;1>/*),pk(@2/<0;1>/*)},pk(@3/<0;1>/*)}) --key @0=xpub6DkFAXWQ2dHxq2vatrt9qyA3bXYU4ToWQwChbf5XB2mSTexCHZCeKS1VZYcPoBd5X8yVcbXFHJR9R8UCVpt82VX1VhR28mCyxUFL4r6KFrF --key @1=xpub6DzhYrFFYQ1HiMDiM388xHnDiRPNdZJFBmmxge3Y1WwChLTmJLfRuhRHqnQCPbTj3fGKTuKFLHzzwpJkp5Dtc3UtlKZKaVZelyqMBXdx6V --key @2=xpub6EGx8sPr9FxpPE1rbZazhQWpMXA3Hf5DYKtZbL7c4BSddzmQktP96UaTvecEkoCZysuaj79GMCFZYT1KKk7Ph2M3Kf5g8B82KZ8T29SKQR --key @3=xpub6E6Z3S5TXJYNJp4U1q3NZ3pCn82i7KXQAKUtNnzLJ3cCdchQeSdFvXemizaHUF7wNwRQAB8mPdoZhGHLiv49cWPtCnoJY3Az3E8JKxH9Mq --fingerprint @0=73c5da0a --fingerprint @1=73c5da0a --fingerprint @2=73c5da0a --fingerprint @3=73c5da0a --path 48'/0'/0'/2' --count 2
bc1p2cycz66cyzltrvy5xd6v6g0jz3m5f9tjzt0aqkx9a5xet23su5rsa4mewu
bc1pa4sqhg0qw69zhqyaw3wv75ke49rwdf9cdc4taah3z5p58lp3zc8se4amnr
note: stdout is watch-only – public keys only, cannot spend
[exit 0]

$ /scratch/code/shibboleth/descriptor-mnemonic/target/release/md address --template tr(@0/<0;1>/*,{pk(@1/<0;1>/*),pk(@2/<0;1>/*)},pk(@3/<0;1>/*)}) --key @0=xpub6DkFAXWQ2dHxq2vatrt9qyA3bXYU4ToWQwChbf5XB2mSTexCHZCeKS1VZYcPoBd5X8yVcbXFHJR9R8UCVpt82VX1VhR28mCyxUFL4r6KFrF --key @1=xpub6DzhYrFFYQ1HiMDiM388xHnDiRPNdZJFBmmxge3Y1WwChLTmJLfRuhRHqnQCPbTj3fGKTuKFLHzzwpJkp5Dtc3UtlKZKaVZelyqMBXdx6V --key @2=xpub6EGx8sPr9FxpPE1rbZazhQWpMXA3Hf5DYKtZbL7c4BSddzmQktP96UaTvecEkoCZysuaj79GMCFZYT1KKk7Ph2M3Kf5g8B82KZ8T29SKQR --key @3=xpub6E6Z3S5TXJYNJp4U1q3NZ3pCn82i7KXQAKUtNnzLJ3cCdchQeSdFvXemizaHUF7wNwRQAB8mPdoZhGHLiv49cWPtCnoJY3Az3E8JKxH9Mq --fingerprint @0=73c5da0a --fingerprint @1=73c5da0a --fingerprint @2=73c5da0a --fingerprint @3=73c5da0a --path 48'/0'/0'/2' --change --count 2
bc1pq67e92udm4tldq48jjdaravtekrw6k7u9apf6alhq2sfe8m9t6hsseycm5
bc1pf0lcpa0n7whukgcapvl022964qyu40h4cq7uc06puphjl28ntf3q9uck2h
note: stdout is watch-only – public keys only, cannot spend
[exit 0]
```

wallet idwallet-descriptor-template-id: 6f58bdf70d6f7d2fa82e8604bdb89317 wallet-policy-id: ade5967c4b3629f43e714498f426582e
receive 0..1bc1p2cycz66cyzltrvy5xd6v6g0jz3m5f9tjzt0aqkx9a5xet23su5rsa4mewu
bc1pa4sqhg0qw69zhqyaw3wv75ke49rwdf9cdc4taah3z5p58lp3zc8se4amnr
change 0..1bc1pq67e92udm4tldq48jjdaravtekrw6k7u9apf6alhq2sfe8m9t6hsseycm5
bc1pf0lcpa0n7whukgcapvl022964qyu40h4cq7uc06puphjl28ntf3q9uck2h

The fingerprints are part of the identity. Measured while building this journey: the same template, the same four xpubs and the same origin, encoded *without* -- fingerprint, gives a 7-chunk set whose wallet-policy-id is 0b3b95a4... instead of ade5967c.... The template id is unchanged either way, because it is key-stable. An operator who omits them gets a card that proves to a different id than their coordinator shows.

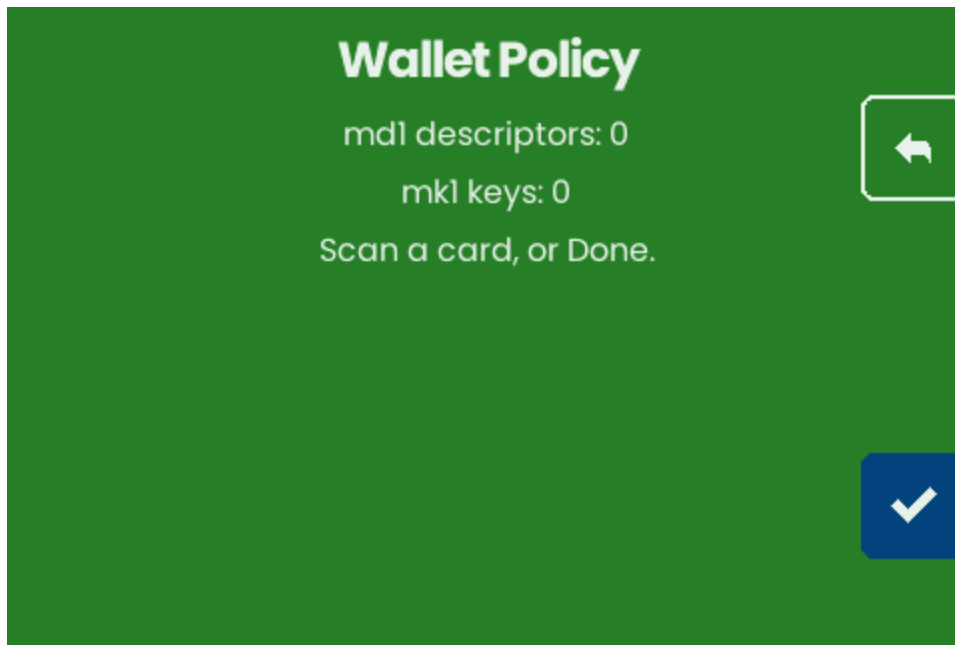
3 · The device



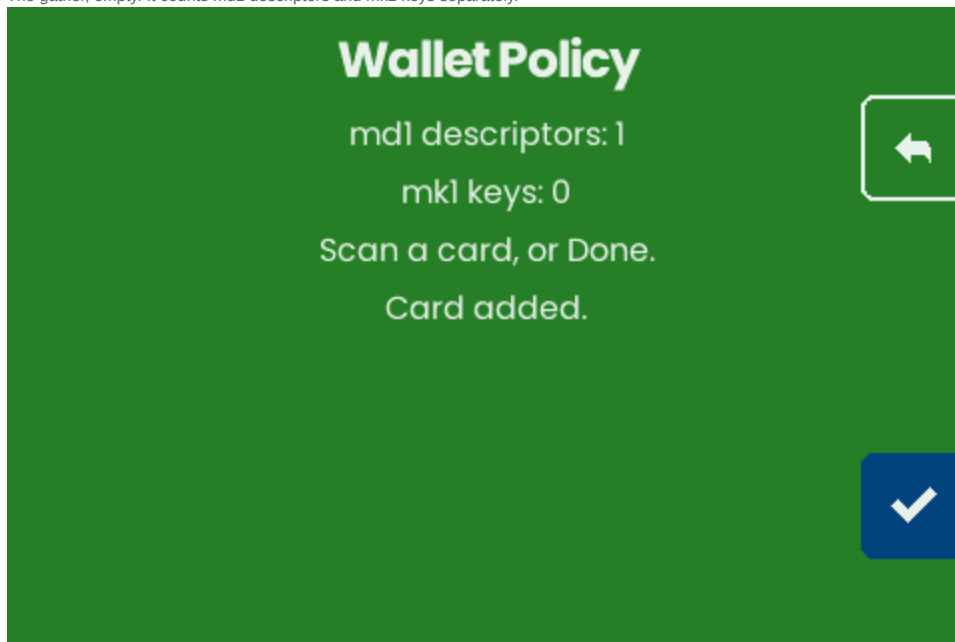
Home. The emulator ships a systemwide payload, so the walk skips its LOAD/SKIP prompt first — this operator's wallet arrives on cards.



Wallet Policy, the eighth entry — beside the other engrave programs, not after the utilities.



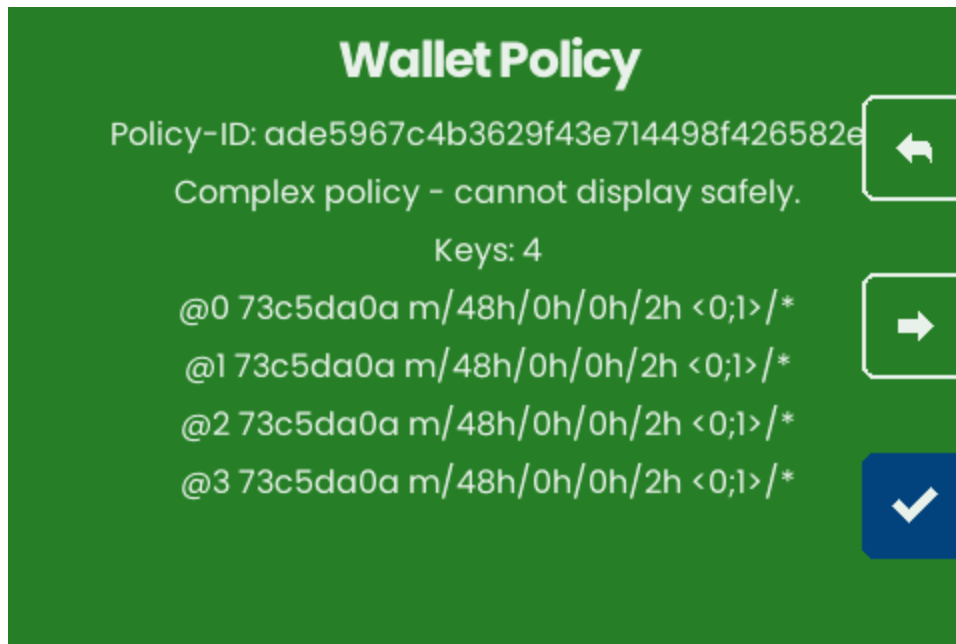
The gather, empty. It counts md1 descriptors and mk1 keys separately.



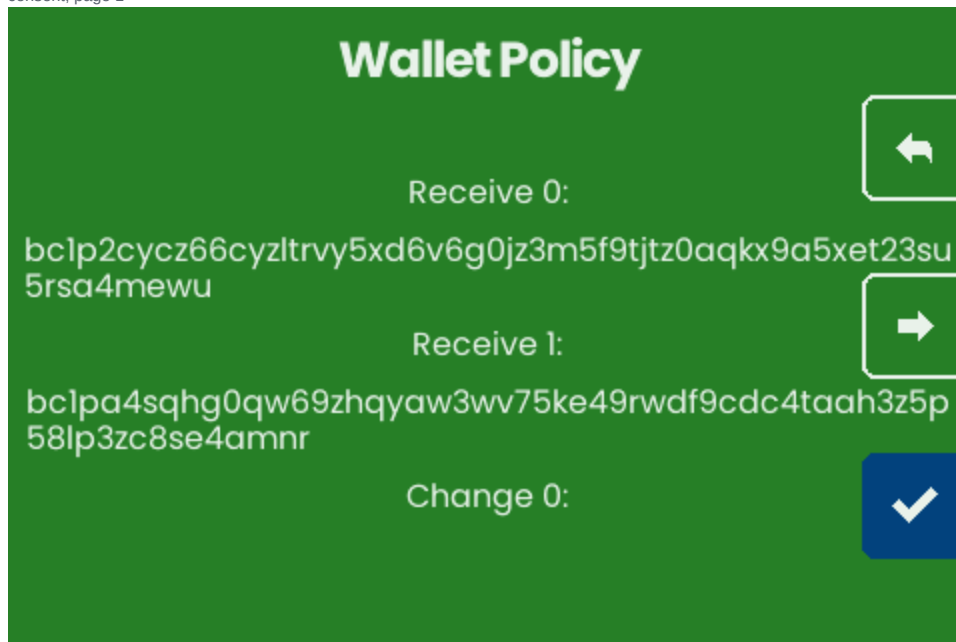
After all 8 chunks cross the reader: ONE card. The tally counts cards, not chunks — a card's intermediate chunks draw nothing, and a dropped one means the set never completes.

4 · The proof, on the consent path

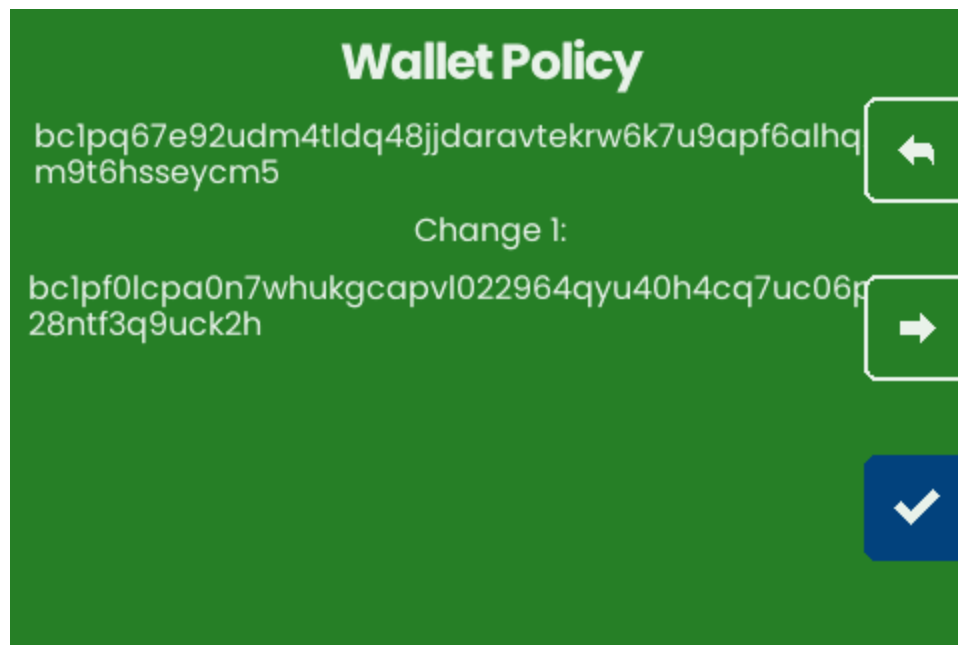
Not beside it. The addresses are lines on the screen the operator has to pass through to engrave, so tapping straight on still shows them.



consent, page 1



consent, page 2



consent, page 3

The id is **named**. Two exist for one wallet — the *key-stable* Template-ID and the *key-dependent* Policy-ID — and they are indistinguishable once rendered as 16 bytes of hex. An operator comparing the wrong one against their coordinator sees a mismatch and reads it as a corrupted backup, so the screen says which it is showing.

5 · What this run actually established

The capture is a **comparison, not a photo shoot**. `shots_walletpolicy.js` is handed the host's id and addresses and throws if the screen disagrees, so these pages cannot show agreement that did not happen.

```
chunks presented8
cards gathered1
consent pages read3
wallet id matchedade5967c4b3629f43e714498f426582e
addresses matchedbc1p2cycz66cyz1trvy5xd6v6g0jz3m5f9tjtz0aqkx9a5xet23su5rsa4mewu
bc1pa4sqhg0qw69zhqyaw3wv75ke49rwdf9cdc4taah3z5p58lp3zc8se4amnr
bc1pq67e92udm4tldq48jjdaravtekrw6k7u9apf6alhq2sfe8m9t6hsseycm5
bc1pf0lcpa0n7whukgcav1022964qyu40h4cq7uc06puphjl28ntf3q9uck2h
```

Two implementations, in two languages, from the same four xpubs: Rust derived these off-device and the firmware derived them again from the cards, and they are the same strings. The negative control was run — corrupting one character of one expected address fails the capture — so the match is a measurement rather than a coincidence of both sides being empty.

Reproducing

```
bash transcript_walletpolicy.sh > transcript_walletpolicy.txt 2>&1
python3 capture_walletpolicy.py      # rebuilds emu.wasm, drives it, writes shots/
python3 build_pdf_walletpolicy.py    # this document
```

What this journey does NOT show

- **No engraving.** The walk stops at consent. Cutting is the Engrave Bundle path, already covered by the pathological journey, and the new thing here is the proof that precedes it.
- **No mk1 key cards.** The program accepts them, and a keyless template gathered alongside its key plates still reports no addresses — that is F-216, deliberately unbuilt because mapping an mk1 to an @N slot has no obvious rule and a wrong mapping would show a *wrong* address as proof.
- **Not every shape.** A tap leaf outside `pk/multi_a/sortedmulti_a` is refused rather than approximated (F-214).